Milestone 2 - Group 4

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For Milestone 2, we further developed the scope and methodological direction of our project on adversarial attacks in class-imbalanced credit-risk modelling. First, we selected the LendingClub accepted-loans dataset as the empirical basis for the project.

Second, we started exploring the dataset to understand its structure and modelling challenges. We inspected the number of observations and features, the distribution of the target variable, missing values, categorical variables, and potentially problematic columns. This initial exploration showed that the original LendingClub dataset contains many, 151 to be exact, different features.

Third, we created a flowchart that structures the project pipeline. The pipeline follows a CRISP-ML(Q)-inspired approach and consists of problem definition, data understanding, data preparation, experimental design, baseline model development, adversarial attack design, adversarial evaluation, defence evaluation, and interpretation/governance. This roadmap helped us clarify how the different parts of the project are connected. In particular, it shows how class-imbalance handling, adversarial robustness, and regulatory interpretability are combined in one experimental framework.

Fourth, we found an existing preprocessed LendingClub notebook that can be used as a practical starting point. This notebook already performs several useful preprocessing steps, such as cleaning variables, handling missing values, encoding categorical variables, scaling features, and training baseline models. However, we also identified that the notebook uses a reduced feature set and is mainly designed for standard loan default prediction. Therefore, we will not copy the notebook directly, but use it as a source of preprocessing inspiration that must be adapted to our research setting.

Finally, we are currently working on refining the novelty and reducing the scope of the project. We decided to focus on default classification, because this better fits the class-imbalance problem. The adversarial attacker will be modeled as a borrower who tries to reduce the predicted probability of default by making small, realistic changes to editable borrower features. This allows us to evaluate whether models trained under stronger class imbalance become more vulnerable to manipulation. The next step is to finalize the feature set, define the imbalance scenarios, implement the baseline models, and specify the adversarial attack constraints.

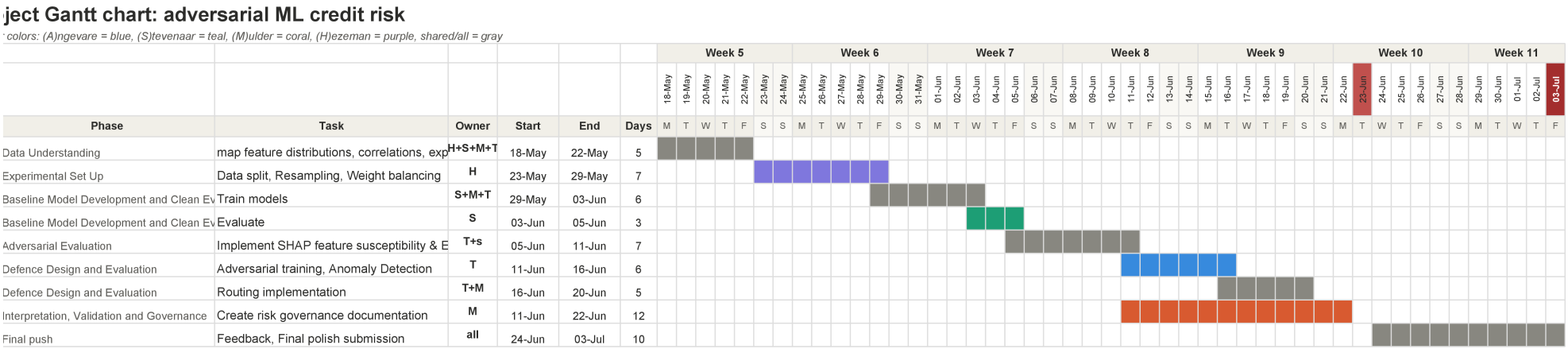


Figure 1: Adversarial machine learning pipeline for credit scoring

AI statement

For this assignment, AI was used for language support, such as checking grammar, spelling, and sentence clarity. In addition, AI was used to brainstorm possible approaches for tackling the assignment. The ideas, arguments, structure, analysis, and final content of this report remain entirely our own.